Homework 4

Problem 4:

a) Code in folder.
Only a total of 5 CSA stages are needed. Each CSA block takes 3 inputs and reduces them to 2 outputs: a sum and a carry, with the carry shifted one bit to the left. In Part 1, the 10 inputs are reduced to 7 outputs using 3 CSA blocks. Part 2 reduces the 7 outputs to 5 using 2 CSA blocks. Part 3 reduces 5 outputs to 4, Part 4 reduces 4 outputs to 3, and Part 5 reduces the final 3 outputs to 2. Once only two values remain, in Part 6 a regular adder combines them to produce the final result.

b) Simulation Results with the sequences: (1,2,3,4,5,6,7,8,9,10) and (3,4,5,6,7,8,9,10).

